UcxLog .UCX record format (unofficial, reverse engineered)

> Status: working specification derived from examined UcxLog normal and contest logs plus CSV exports. This is **not** an official UcxLog format description.

Container

The examined .UCX files are flat arrays of fixed-size QSO records:

```
record_size = 364 bytes
record_count = file_size / 364
record_offset(n) = n * 364
```

No global file header was observed in the examined files. A file whose size is not divisible by 364 should be treated as unexpected/corrupt/unsupported by the current reader.

All identified multi-byte integer values are little-endian.

Most fixed text slots use a Pascal/Delphi-like short-string representation:

```
[length byte][fixed-capacity character area]
```

The reader validates that the stored length does not exceed the slot capacity.

Confidence markers

- **Confirmed**: directly matched across records and/or CSV/UcxLog UI.
- **Structural**: fixed slot/structure is clear, semantic meaning not fully proven.
- **Unknown**: intentionally not interpreted.

Record map

Offset	Size	Encoding	Meaning	Confidence
0	1	byte	1 in all examined normal-log records, 0 in examined contest records	Structural
1	1	byte	day	Confirmed
2	1	byte	month	Confirmed
3	2	UInt16 LE	year	Confirmed
5	1	byte	start minute	Confirmed

Offset	Size	Encoding	Meaning	Confidence
6	1	byte	start hour	Confirmed
7	1	byte	end minute	Confirmed
8	1	byte	end hour	Confirmed
9	16	ShortString[15]	callsign	Confirmed
25	4	ShortString[3]	sent RST	Confirmed
29	20	ShortString[19]	sent exchange	Confirmed
49	4	ShortString[3]	received RST	Confirmed
53	20	ShortString[19]	received exchange	Confirmed
73	1	enum	band code	Confirmed
74	1	enum	mode code	Confirmed
75	23	ShortString[22]	QTH	Confirmed
98	13	ShortString[12]	name	Confirmed
111	40	ShortString[39]	remarks	Confirmed
151	1	bitfield	print/QSL status #1	Partially confirmed
152	8	ShortString[7]	country / DXCC-style country prefix used by UcxLog	Confirmed
160	1	enum	continent	Confirmed for EU/AS/AF/NA/SA/OC
161	1	bitfield	status #2	Partially confirmed
162	9	ShortString[8]	unknown string	Structural
171	4	bytes	unknown	Unknown
175	8	UInt64 LE	exact frequency in Hz	Confirmed
183	12	ShortString[11]	contest multiplier 1	Confirmed
195	12	ShortString[11]	contest multiplier 2	Structural/confirmed slot
207	4	UInt32 LE	contest QSO points	Confirmed
211	9	ShortString[8]	Maidenhead locator	Confirmed
220	2	UInt16 LE	IOTA number	Confirmed
222	1	enum	IOTA continent	Confirmed
223	9	ShortString[8]	district	Confirmed
232	1	byte	CQ zone	Confirmed

Offset	Size	Encoding	Meaning	Confidence
233	1	byte	ITU zone	Confirmed
234	9	ShortString[8]	QSL manager / via	Confirmed
243	1	byte	unknown	Unknown
244	1	boolean	CBA - QSL direct / sort forward for printing	Confirmed
245	1	boolean	Ok - manager verified/safe	Confirmed
246	8	IEEE-754 double LE	Delphi TDateTime paper-QSL send timestamp	Confirmed
254	1	byte	unknown	Unknown
255	6	ShortString[5]	unknown string	Structural
261	11	ShortString[10]	unknown string	Structural
272	12	ShortString[11]	Award 1 abbreviation	Confirmed
284	12	ShortString[11]	Count 1 / arbitrary award value text	Confirmed
296	12	ShortString[11]	Award 2 abbreviation	Confirmed structurally/semantically from UcxLog help
308	12	ShortString[11]	Count 2 / arbitrary award value text	Confirmed structurally/semantically from UcxLog help
320	22	bytes	internal metadata	Unknown
342	20	bytes	zero in all examined samples	Unknown/reserved
362	2	UInt16 LE	changing trailer/check value; algorithm unknown	Unknown

Band code at offset 73

Observed/derived mapping:

Code	Band	Evidence
0	160 m	observed
1	80 m	observed
2	40 m	observed
3	30 m	derived from UcxLog UI order
4	20 m	observed

Code	Band	Evidence
5	17 m	observed
6	15 m	observed
7	12 m	observed
8	10 m	observed
9	6 m	observed
10	2 m	observed
11	70 cm	derived from UI order
12	23 cm	derived from UI order
13	13 cm	derived from UI order
14	3400 MHz	derived from UI order
15	5600 MHz	derived from UI order
16	10 GHz	observed (microwave/SAT record)

UcxLog also offers `unbestimmt` and `Rest`; their numeric values have not been proven.

The raw numeric band code is therefore retained even when a friendly enum name is known.

Mode code at offset 74

Controlled test records established:

Code	Mode
0	CW
1	SSB
2	RTTY
3	AM
4	FM
5	PSK31
7	PKT
12	SAT
30	FT8

Other UcxLog modes are not yet mapped. The raw mode code is always retained.

QSL/print status byte at offset 151

Confirmed interpretation:

```
bits 0..1  print mark
             0 = empty
             1 = P  paper QSL should be printed/sent
             2 = n  do not send this time
             3 = N  never send

bit 3  0x08 = paper QSL sent
bit 4  0x10 = unknown (do not label as LoTW upload without a hard proof)
bit 5  0x20 = paper QSL received
bit 6  0x40 = eQSL confirmed
bit 7  0x80 = LoTW confirmed
```

Bit 2 (`0x04`) remains unknown.

Status byte at offset 161

```
0x20 = ClubLog confirmed     [confirmed]
0x80 = unknown               [intentionally left unknown]
```

No QRZ/ClubLog upload interpretation is currently committed to the format specification.

QSL send timestamp at 246

The 8 bytes decode as a little-endian IEEE-754 double compatible with Delphi `TDateTime`, using 1899-12-30 as day zero for the modern positive values observed.

Important: a stale non-zero timestamp can remain in a record even if no paper QSL is currently marked as sent. Treat the timestamp as valid only when `(status151 & 0x08) != 0`.

IOTA encoding

```
220..221 = UInt16 LE numeric part
222      = continent enum
```

Observed IOTA continent enum:

```
0 EU
1 AS
2 AF
3 NA
4 SA
```

```
5 OC
6 AN
```

QSO continent at offset 160

Observed:

```
0 unknown/not set
1 EU
2 AS
3 AF
4 NA
5 SA
6 OC
```

Antarctica was not present in the analyzed samples and is intentionally not guessed.

Contest data philosophy

The UCX record stores generic strings and values rather than contest-specific semantics:

- Sent RST
- Sent exchange
- Received RST
- Received exchange
- Multiplier 1
- Multiplier 2
- Points

For example, an exchange may contain a serial number, locator, zone, district, HQ abbreviation or another contest-defined value. The parser should not reinterpret it.

Contest QSOs must remain normal statistical QSOs; the contest marker/fields are additional metadata, not a reason to exclude them from band/DXCC/locator/award statistics.

QTC files

A tested `.QTC` file was a separate fixed-width line-oriented text format and is not part of the 364-byte UCX record. QTC support is deliberately outside the initial parser scope.

Bestätigte Online-Statusbits

- Offset 151, `0x10`: zu LoTW hochgeladen
- Offset 151, `0x40`: eQSL bestätigt
- Offset 151, `0x80`: LoTW bestätigt
- Offset 161, `0x20`: ClubLog bestätigt
- Offset 161, `0x80`: zu ClubLog hochgeladen

QRZ.com wird von UcxLog als Uploadziel genutzt; im 364-Byte-QSO-Record wurde bislang kein persistenter QRZ-Uploadstatus nachgewiesen.

Locator precision

UcxLog accepts at most 8 Maidenhead characters. The UCX record stores the locator length at offset 211 and up to 8 characters at offsets 212-219. An 8-character test locator (`JO40FG12`) was stored without truncation.